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25 years of factorial surveys in sociology: A review

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ABSTRACT

The aim of the factorial survey approach is to uncover the social and individual structures of human judgements of social objects. By having respondents evaluate samples of vignettes (fictive descriptions), in which several factors describing the object of interest are simultaneously manipulated, this approach has a number of advantages over traditional social survey research. The aim of the present article is to provide an easy-to-follow overview of the various ways in which the approach has been applied within sociology between 1982 and 2006. The review, which is based on 106 articles published in central sociology journals, is organised into three different sections: "factorial survey applications", "factorial survey designs" and "factorial survey analyses".

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1. Introduction

More than three decades ago, the first study based on the *factorial survey approach*—a method intended for the investigation of human judgements—was published (Rossi et al., 1974). The publication of this article ("Measuring Household Social Standing") followed a period of several decades during which the accredited founder of this approach, Peter Henry Rossi, had been pondering the numerous benefits of integrating the principles of the experiment with those of the social survey in a single method for data collection. Although the late 1970s saw a number of extensions and applications of this method (e.g. Sampson and Rossi, 1975; Jasso and Rossi, 1977), it was not until the publication of *Measuring Social Judgments* in 1982 (edited by Peter Rossi and Steven Nock), that the principles of the approach were carefully and systematically presented to the wider research community. In the introductory chapter of this milestone volume, the authors stated that the aim of the factorial survey approach was to make it possible to determine the underlying principles behind human judgements (or evaluations) of social objects (Rossi and Anderson, 1982). By "judgements" of "objects", they meant the assigning of values to "actions, objects, other persons, other groups, institutions, ideas, and so on" (Rossi and Anderson, 1982, p. 15). Rossi and Anderson (1982) proceeded from the assumption that judgements of social objects, however idiosyncratic they may seem, are socially and individually structured, and that it is the task of the social scientist to uncover these structures. Indeed, considering the long tradition within the social sciences of research on attitudes (opinions, beliefs; cf. DeLamater, 1992), it may be claimed that one of the most important undertakings for social scientists in general, and for sociologists in particular, is that of determining the social causes and social outcomes of human judgements.

In an effort to come to terms with the limitations of traditional social survey research—where human judgements are derived either from respondents' answers to direct questions or from their judgements of statements (cf. Alexander and Becker, 1978)—Rossi and Anderson (1982) proposed that researchers use standardized vignettes (fictive descriptions), in which selected characteristics describing the objects to be judged by respondents are simultaneously manipulated. By comparison with traditional social survey research, this approach—which may be described as quasi-experimental—presents a number of advantages. First, because it involves presenting respondents with concrete and detailed descriptions in which several different factors believed to influence the judgement being studied are systematically varied, the approach is better

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suited to studying the contexts and conditions affecting judgements. To take one example, while results from conventional surveys have repeatedly suggested that the general population of the United States is highly supportive of the death penalty for people convicted of murder (in October 2007, for example, the level of support was 69%, see Gallup Poll Online, 2007), results from a factorial survey of capital punishment preferences (Boots et al., 2003) showed that when presented with realistic homicide scenarios in which the circumstances of the crime and the characteristics of the offender were described in detail, respondents proved rather unwilling to recommend a sentence of death. In addition, the results revealed that the offender's age, mental capacity and prior arrest record had a powerful influence on respondents' willingness to recommend the death penalty. Second, because respondents in factorial survey studies are most probably not fully attentive to the manipulation—i.e. the controlled variation—of the different elements of the vignettes, the judgements detected by such a design may arguably be less subject to social desirability bias than those obtained using conventional survey items (Alexander and Becker, 1978). In an examination of the effect of the race of strangers encountered in public settings on the fear evoked by such encounters, for example, St. John and Heald-Moore (1995) found that white Americans were more afraid of meeting black strangers than white strangers, irrespective both of a number of other characteristics of the strangers encountered and of the specific settings in which the meetings took place. Finally, because people are not always aware of the influences certain factors have on their judgements—and are thus incapable of explicating such influences when asked about them—it has been argued that the factorial survey approach is particularly appropriate when researchers want to study the *actual* determinants—or combinations of determinants—of human judgements (Alexander and Becker, 1978). In one of the first FSA studies using professionals as respondents, for example, Jasso (1988) identified a number of principles guiding the judgements of staff members (at the U.S. Select Commission on Immigration and Refugee Policy) about the relative desirability of different types of visa applicants as immigrants to the United States. One result from the study was that no two staff members employed a similar set of principles for judging the desirability of visa applicants with different personal, kinship, market and contextual characteristics.

Despite the above named advantages associated with the use of the factorial survey approach in the study of human judgement, and despite the fact that the approach has now been around for over a quarter of a century, my own overriding impression—having presented my own research at several research conferences and seminars—is that most sociologists have never heard of it. This may of course be due to the fact that its inclusion in textbooks on research methods for the social sciences is far from standard (for an exception, see Fink, 2002). The lack of an up to date literature to guide factorial survey studies was recently partially filled by the publication of a comprehensive article by Guillermina Jasso (2006) in *Sociological Methods and Research*. However, since Jasso's article focused on formulating a unified framework for the study of particular forms of judgements—that is positive beliefs and normative judgements—the article does not cover one of the three main ways in which the approach has to date been applied in sociology, namely to study own intentions. In addition, due to the sophisticated technical language used by the author, social scientists who lack prior advanced knowledge of statistical analysis may find it difficult to fully assimilate the contents of Jasso's article. Thus, in line with the wish expressed by Rossi (1979, p. 186) , that the factorial survey approach become "a part of standard repertory for social researchers", the objective of this article is that of providing an overview of the various ways in which the approach has been applied within the discipline of sociology over a period of 25 years, and in a form that can easily be followed by readers with no advanced knowledge of statistics. It is not my intention to critically evaluate the way the factorial survey approach has been employed by various researchers, but rather to compile and categorize information from a large number of studies in order to shed light on the various possibilities afforded by the method. This article is structured as follows: To begin with, I will provide a brief introduction of the factorial survey approach, and present details of the search strategies and criteria employed to identify articles for inclusion in the review. I will then move on to review selected aspects of the 106 articles so identified, incorporated under the general headings of "factorial survey applications", "factorial survey designs" and "factorial survey analyses".

1.1. An introduction to the factorial survey approach

The main components of factorial surveys are the vignettes (also referred to as *factorial objects*, see Rossi and Anderson, 1982) that are judged by the respondents. For respondents in an FSA study, vignettes are fictive descriptions of social objects, constituted either of people or of social situations (cf. Alexander and Becker, 1978). For the researchers in charge of the study, the vignettes represent different combinations of *levels* (values) of various *dimensions* (variables), which are included on account of their presumed relevance as determinants of the judgement of interest. Below is an example of a vignette, borrowed from a study concerned with the underlying causes of racial residential segregation (Emerson et al., 2001). This study was part of a national survey of attitudes and values among 1663 white Americans. Fig. 1 presents the introductory statement used by the authors, followed by a description of a vignette consisting of six dimensions —the dimension levels, one of which was chosen for each vignette, are displayed in brackets—and a rating task.

The factorial survey approach has often been referred to as a "hybrid" technique which combines the factor orthogonality—i.e. perfect non-association between dimensions—that is characteristic of the balanced multivariate experimental design with the rich detail and complexity that may be obtained using sample survey procedures (Rossi and Anderson, 1982). By letting each of the dimensions included in the vignettes vary independently with respect to its levels, factor orthogonality is obtained across dimensions in the vignette *universe*, which consists of all possible combinations of dimension levels. The vignette universe of the design described above consists of the (Cartesian) product of the levels attached to the six dimensions, in this case $3 \times 20 \times 3 \times 3 \times 3 \times 3 = 4860$ different vignettes. By having respondents judge a randomly or sys-

"Imagine that you are looking for a new house and that you have two school-aged children. You find a house that you like much better than any other house – it has everything that you'd been looking for, it is close to work, and it is within your price range."

"Checking on the neighborhood, you find that the public schools are of [low, medium, high] quality, the neighborhood is [5-100%] [Asian, black, Hispanic], property values are [declining, stable, increasing], the other homes in the neighborhood are of [lower value than, equal value to, higher value than] the home you are considering, and the crime rate is [low, average, high].

How likely or unlikely do you think it is that you would buy this home?"

(Responses range from very unlikely [1] to very likely [6].)

Fig. 1. Example of a vignette (Emerson et al., 2001, p. 925).

tematically drawn *sample* from the vignette universe, instead of having them judge all possible combinations of dimension levels (which is the case in factorial experiments, cf. Rossi and Nock, 1982), it is possible to include a large number of vignette dimensions and levels in the design, thereby "enhancing the resemblance between real and experimental worlds" (Rossi and Anderson, 1982, p. 16). The presence of approximate factor orthogonality in the samples of vignettes that are judged by the respondents makes it possible for the researcher to disentangle the unique effect of dimensions that are normally very highly correlated. For example, the primary aim for Emerson and his colleagues (2001) was to investigate whether the racial composition of the neighbourhood had an effect on white Americans' residential preferences independent of variables that may serve as its proxies, such as housing values and crime rates, for example. While the high correlation found in the United States between these variables had made it difficult to use behavioural data to infer the unique contribution of the racial composition of neighbourhoods on residential preferences (cf. Emerson et al., 2001), the use of FSA vignettes—in which the above-described variables were uncorrelated—allowed Emerson and his colleagues to disentangle the effects of the different variables on the outcome. The other side of the coin as far as the factor orthogonality characteristic is concerned is that some respondents may be presented with descriptions which combine elements that seldom co-occur in the real world.[1] In a critical comment on Alves and Rossi's factorial survey article published in the *American Journal of Sociology* in 1978, Faia (1980) argued that studies which make use of such "unrealistic" fictive descriptions unavoidably generate unrealistic results, since the respondents, when presented with unusual combinations of dimension levels, may start making judgements that do not accurately reflect the principles that they would have used had the vignettes been realistic. In response, Rossi and Alves (1980) argued that it is the judgements made of statistically infrequent combinations of characteristics that most clearly reveal the respondents' true principles of judgement. However, because respondents' judgements of fictitious objects cannot be straightforwardly generalized to their judgements of real objects, Rossi (1979) has also insisted that authors of factorial survey studies pay continuous attention to the issue of the realism of their results.

1.2. Review inclusion criteria

It is never either easy or unproblematic to define the criteria that determine which publications should be included in a review article. This review comprises all factorial survey studies written in English and published in "core"[2] or "priority"[3] sociology[4] journals (as classified in the "Sociological Abstracts" database) between 1982 and 2006 (Sociological Abstracts, 2008a). Because the volume edited by Rossi and Nock (1982) may be viewed as a milestone, and because it signals a first step towards this method's maturity, the year of its publication was used as the lower bound for the selection of articles for the review. The choice of excluding FSA studies published in monographs (e.g. Rossi and Berk, 1997) and articles appearing in journals that are either classified in Sociological Abstracts as "selective"[5] (e.g. Rossi et al., 1985) or that do not appear on the serials

[1] However, factor orthogonality does not inevitably produce "unrealistic" vignettes. For example, in the study by Jasso (1988), all the descriptions making up the vignette population resemble the real-world visa applicants discussed by immigration policymakers.

[2] "Core journals are published by sociological associations, groups, faculties, and institutions, and/or have the term "sociology" in their titles" (Sociological Abstracts, 2008a).

[3] "Priority journals are drawn from related social sciences, e.g., anthropology, communication, education, medicine, political science, which consistently address topics of sociological relevance, and in which sociologists regularly publish" (Sociological Abstracts, 2008a).

[4] The choice of restricting this review to the discipline of sociology, despite the fact that the factorial survey approach has also been used in other disciplines, such as economics (e.g. Goodman, 1989) and medicine (e.g. Brown et al., 1997), was grounded in the fact that the method was first introduced by a sociologist (Peter Rossi). It could also be noted that this method has been referred to as "sociological" factorial survey analysis (Goodman, 1989, p. 58).

[5] "Selective sources include journals from related social sciences that occasionally publish works by sociologists" (Sociological Abstracts, 2008a).

Table 1
Subdisciplines of sociology (n = 106).

	Frequency
Crime, law, and deviance[a]	49
Family and social welfare[b]	18
Social differentiation[b]	12
Sociology of health and medicine[c]	5
Organizations, occupations, and work[a]	5
Urban sociology[c]	5
Sociological theory[c]	5
Political sociology[c]	2
Ethnic relations[c]	2
Sex and gender[a]	1
Religion[a]	1
Sociology of sports[c]	1

[a] American Sociological Association section.
[b] Sociological Abstracts subject.
[c] Sociological Abstracts subdiscipline.

source list (e.g. Wallander and Blomqvist, 2005), was made for reasons of practicality. Although this review does not cover all factorial survey studies qualifying as "sociology", it includes all studies published in journals considered to be central within the discipline of sociology. Because these journals together not only constitute the most important forum for the publication of research carried out by sociologists, but also implicitly pass on the notion of what has been counted (and what may be counted) as sociology to future generations of sociologists, I believe that a sample such as the one chosen for this review is satisfactory. It must be emphasized, however, that the conclusions drawn in the article are based on a selection of studies and that the excluded research contains some notable highlights in the history of the factorial survey, such as Rossi and Rossi (1990) and Rossi and Berk (1997).

During the autumn of 2007 final searches for articles were conducted in multiple databases and in the following manner. Since the review proceeds from the works edited by Rossi and Nock in "Measuring Social Judgments" (1982), I started by searching for articles that included references to this book. From a total of 193 matches in the "Social Sciences Citation Index", I excluded (1) matches consisting of articles published in journals that were not defined as core or priority on the serials source list in the Sociological Abstracts database; and (2) matches that did not in themselves include factorial survey applications to substantive areas of research. Among the articles not included in the review were those which focused on a discussion of the method in itself (e.g. Jasso, 2006), for example, and those that were described by the authors as factorial surveys, although they in fact constituted examples of factorial experiments (e.g. Baker, 1983). Following the exclusion of articles on the basis of these criteria, 76 articles remained. Thereafter, since the authors of factorial survey articles may not necessarily refer to the book from 1982, I searched several social sciences databases for articles in which the terms "factorial survey" appeared together anywhere in the text, as well as articles where the terms "Rossi" and "vignette" ("vignette*") appeared anywhere in the text.[6] Databases generating further material for the review included "Sociological Abstracts", "PsychINFO", "Criminal Justice Abstracts", "Academic Search Premier", and "JSTOR". In total, 106 articles were elected for inclusion in the review, representing 92 original studies.[7]

2. Factorial survey applications

2.1. Factorial surveys in subdisciplines of sociology

In order to further investigate the use of the factorial survey approach in sociology, it is useful to classify the studies into subdisciplines. The categorization of articles was made primarily on the basis of information provided in the articles' abstracts regarding the specific judgement being studied. Table 1 shows the distribution of FSA articles over subdisciplines of sociology.[8] As may be noted, a large proportion of the publications reviewed focused on subjects related to the field of crime, law and deviance. Within this broad area, the approach has frequently been used in the study of amongst other things public punishment preferences (n = 12; e.g. Boots et al., 2003), sexual offending (n = 7; e.g. Rossi and Weber-Burdin, 1983), and corporate crime (n = 5; e.g. Elis and Simpson, 1995). The subdiscipline category labelled family and social welfare includes a number of studies investigating the judgements of welfare professionals, most of them in relation to child abuse (n = 5; e.g. O'Toole et al., 1993) and aging (n = 3; e.g. Degenholtz et al., 1999). This category also includes eight studies focusing on beliefs about intergenerational responsibilities (e.g. Ganong and Coleman, 2005). When it comes to the subject of social differentiation, the factorial survey approach has been used to examine perceptions of fairness in relation to earnings (n = 3; e.g. Jasso and Web-

[6] Occasionally, the vignettes employed in factorial survey studies have been described as "Rossi vignettes" (Anderson, Harris and Miller, 1983, p. 238).
[7] The total number of unique samples—or groups of samples—described in the articles defines the number of original studies in the review.
[8] My list of subdisciplines is comprised of a combination of subjects (Sociological Abstracts, 2008b) and subdisciplines listed in Sociological Abstracts (2006) and of sections included in the American Sociological Association (2008).

Table 2
Forms of judgement (n = 132).

	Frequency
Normative judgement	62
Recommendation	19
Responsibility	15
Seriousness	14
Fairness	12
Other	2
Positive belief	26
Definition	17
Estimation of outcome	6
Other	3
Own (intended) actions, feelings and thoughts	44
Actions	28
Feelings	8
Thoughts	8

ster, 1999) for example, and perceptions of the poor (n = 3; e.g. Will, 1993). The focus of the studies classified in the health and medicine category varies, with one study focusing on popular conceptions of mental illness (Thurman et al., 1988) for example, and another comparing the disease-related stigma of AIDS and cancer (Schulte, 2002). Diversity in subject matter also characterises the group of studies classified as belonging to the subdiscipline labelled organizations, occupations and work, which includes an investigation of the selection behaviour of employers (de Wolf and van der Velden, 2001) for example, and a study of managers' perceptions of commitment to the organization (Porter, 2001). By contrast, the urban sociology articles have had a uniform focus on modelling public evaluations of neighbourhoods (n = 4; e.g. Shlay and DiGregorio, 1985).

Naturally, all of the studies included in the review have to some degree been informed by sociological theory. There is however a group of studies in which the specific judgement measured by the vignette ratings did not constitute the focus of the study, but only served as an example employed to illuminate aspects of more general sociological theory. Opp (2002), for example, explored the existence of an informal non-smoking norm in various contexts and circumstances, with the specific goal of investigating whether norms emerge by human design or as the unintended consequences of human action. The remaining subcategories comprise articles investigating a broad range of subjects including norms for political action (Jasso and Opp, 1997), attitudes towards immigration (Short and Magaña, 2002), occupational choice and the marriage market (Badgett and Folbre, 2003), the effects of biases on pastor placements (Schreckengost, 1987) and sports officials' judgements of spectator behaviour (Ward et al., 2002).

2.2. Forms of judgements in factorial surveys

Rossi and Anderson (1982) maintained that the factorial survey approach should be used in order to measure social judgments. However, besides giving examples of objects that may be regarded as "social", and arguing that these social objects "typically. . .have many traits on which they vary" (Rossi and Anderson, 1982, p. 18), the authors did not provide any clues as to what they actually meant by social judgements. In an effort to obtain a more thorough understanding of what forms of judgements might be investigated in factorial surveys, we may instead turn to Jasso, who claimed that this particular approach is optimal for investigating what she designates "positive beliefs" and "normative judgements" (Jasso, 2006, pp. 335–336). While positive beliefs represent individuals' beliefs about how something "is", normative judgements refer to individuals' judgements concerning how something "ought to be". Thus, in rating the fictive descriptions employed in the factorial survey approach, the respondents might act either as "lay scientists" or as "lay judges" (Jasso, 2006, pp. 335–336). In an effort to categorize the judgements studied in the articles included in the current review, I made use of Jasso's distinction between positive beliefs and normative judgements. However, I soon realized that there was a group of judgements which did not easily fall into either of the two categories. These comprised individuals' estimations of their own thoughts, feelings and intentions to act. On the one hand, the respondents' estimations about what they *would* think, what they *would* feel and how they *would* act (in response to the situations described in the vignettes) could be thought of as individuals' positive beliefs about themselves. However, individuals' estimations of their own thoughts, feelings and intentions to act could also be interpreted as representing their normative judgements about how the actor(s) described in the scenarios *should* think, feel or act. Because the intentions of the FSA articles' authors—as regards this difference in connotations—were not always obvious, I chose to form a separate category for these judgements. As will be noted below, I found it useful to further categorize the judgements into subcategories[9], which are displayed in Table 2.[10]

Table 2 reveals that the form of judgement most frequently addressed in the reviewed FSA studies is normative. In addition, it shows that the most common form of normative judgement is the recommendation, such as a sentencing rec-

[9] Irrespective of the number of similar judgements examined in an article, only one match per judgement subcategory is counted.
[10] Because the authors of some articles have studied several forms of judgement, the sum of judgements is larger than the sum of articles.

Table 3
Details about the respondents ($n = 92$).

Country	Frequency	Respondents	Frequency
USA	76	General population	40
Israel	4	Students	20
Canada	3	Professionals	16
Germany	3	Several samples	12
The Netherlands	2	Volunteers	4
United Kingdom	1		
Cross-national	3		

ommendation for example (e.g. Cochran et al., 2003). In 15 cases, the authors investigated judgements of actors' responsibilities. The numerous studies by Ganong and colleagues on the subject of intergenerational obligations (e.g. Ganong and Coleman, 2006a), make up a considerable portion of the articles included in this subcategory. Judgements dealing with the seriousness of acts or events have also frequently been studied by sociologists, as for example in the study by Hembroff (1987), who explored public judgements about the seriousness of crimes in different contexts. Finally, normative judgements may also be expressed as judgements of fairness, as was the case in Jasso's articles (e.g. Jasso, 1994) on the fairness of earnings, for example. Further, the review includes 26 articles which address the positive beliefs of individuals. The largest subcategory of positive beliefs comprises definitions of social objects, including for example public beliefs about sexual harassment (e.g. Reilly et al., 1982), and professional beliefs regarding child abuse (e.g. O'Toole et al., 1997). Another example of positive beliefs is found in the estimation of an outcome, following the events or actions described in the vignette. Roberts and White (1986), for example, investigated individuals' estimations of the likelihood that an offender would recidivate. Finally, individuals' estimations of their own intended actions, thoughts and feelings were represented in 44 cases. Several authors have focused on individuals' estimations of their own actions, as in the case of Thurman et al. (1993) for example, who explored individuals' decisions to drink and drive. Most FSA studies on the subject of individuals' thoughts about a social object were concerned with preferences, as in the article by Pampel et al. (1984) for example, about the attractiveness of locations as places to live following retirement. Finally, the articles investigating individuals' feelings in relation to a social object included Warr's (1990) study which focused on people's fear of becoming the victims of crime.

2.3. Respondents in factorial surveys

Since it is the researchers who choose the respondents for a study, the question of whose judgements have been studied by factorial surveys is partly determined by the geographical location of the researchers. As this article reviews only those publications that have appeared in selected English-speaking journals, it may not provide a good-enough picture of the extent to which the factorial survey approach has spread to research communities of sociologists in different countries. I know for example that a number of FSA articles have appeared in German journals (e.g. Beck and Opp, 2001; Mäs et al., 2005). It is notable however that 76 of the 92 unique samples (or groups of samples) employed in the 106 articles were collected in the United States alone (see Table 3). The remaining single-country samples were from Israel (e.g. Herzog, 2003), Canada (e.g. Roberts and White, 1986), Germany (e.g. Opp, 2002), the Netherlands (e.g. de Wolf and van der Velden, 2001) and the UK (Evans, 1997). Two of the three studies employing samples from more than one country explicitly compared the judgements of individuals across countries (Russia, Japan and the United States in Sanders et al. (1998), for example; the United Kingdom and the United States in Herman and Heimovics, 1994).

As can also be seen from Table 3, most studies were concerned with judgements made by members of the general population. In 20 studies, researchers relied on student samples. There are a variety of reasons why students have been utilized as respondents in factorial surveys. While students actually comprised the target population of interest in some of these studies (e.g. Carlson, 1996), other studies sampled students for reasons of accessibility (e.g. Thurman et al., 1988). Further, the factorial survey approach has frequently been used in order to study "professional" judgements and beliefs.[11] In the studies reviewed, the occupational groups of interest included behavioural change professionals (Love et al., 1990), case managers (e.g. Corazzini-Gomez, 2002), juvenile court judges (Applegate et al., 2000), different groups of welfare staff (e.g. Rindfleisch and Bean, 1988), employers (de Wolf and van der Velden 2001), managers (e.g. Morrill et al., 1997), nurses (e.g. O'Toole et al, 1993), pastors (Schreckengost, 1987), police officers (Son et al., 1998), sports officials (Ward et al., 2002), staff at the U.S. Select Commission (Jasso, 1988), teachers (e.g. Webster et al., 2005), and the top staff aides of U.S. Senators and Representatives (Cahan, 1996).[12]

[11] In this article, "professional" judgements comprise the judgements made by those who were selected as respondents because of their membership of a specific occupation. This does not include respondents (e.g. university staff, see Reilly, Carpenter, Dull and Bartlett, 1982) who were treated as part of a convenience sample.

[12] Peter Rossi was instrumental in introducing the factorial survey approach to the study of professional judgements (e.g. Rossi, Simpson and Miller, 1985).

Table 4
Rating tasks (n = 123).

	Frequency
Ordered categories	57
Number matching	14
Unordered categories	18
Amount	9
Probability	5
Multiple items	8
Open-ended task	7
Other	5

Interestingly, the employment of several samples did not necessarily imply that the authors had a specific interest in comparing the judgements of individuals across samples. The comparison of judgements across samples was treated as central in only 5 of the 12 studies of this kind included in the review, with one example being the study by Baron et al. (2001a), in which the authors compare the conflict management styles of young males living "on the street" with those of young males in the general population. Finally, the group of samples that have been characterised as comprising "volunteers" is something of a mixed bunch, including for example visitors to a shopping mall (Anderson et al., 1983) and elderly residents in selected cities (Hechter et al., 1999).

3. Factorial survey designs

3.1. Rating tasks in factorial surveys

In a factorial survey, the judgement to be studied is operationalised into a rating task, which is appended to the vignettes. Since the respondents' ratings of the vignettes generate the study's outcome variable, the measurement properties of the rating task determine what forms of techniques may be used in subsequent analysis.[13] According to Jasso (2006), rating tasks in FSA studies may be constructed so as to define an amount, a probability, a set of unordered categories or a set of ordered categories. Table 4 shows that by far the most frequently used rating task in the factorial surveys reviewed in the present context consisted of a set of ordered categories.[14] A typical example of such a rating task is the one used by Herzog in his article from 2004b, which presents findings from a study in which the respondents were asked to judge the seriousness of murder scenarios on a scale from 1 ("not serious at all") to 11 ("very serious"). A further analysis of the 57 FSA articles which used ordered categories as rating tasks shows that the median and modal number of categories marked out on the scales used were 10 and 11, respectively (n for the mode = 17), whereas the minimum and maximum numbers of categories were 4 (n = 5; e.g. St. John and Heald-Moore, 1996) and 21 (Schreckengost, 1987).

One weakness with scales consisting of limited sets of ordered categories is that they force respondents to "stay within a set range and to round off their judgements" (Jasso and Rossi, 1977, p. 643). An alternative is to use a form of scale known as "number matching" (cf. Jasso, 2006), which involves having the respondent match a number to a point (or mark a point) on a response-variable continuum. Fourteen of the articles included in this review (e.g. Liebig, 2001) used number matching scales for the vignette ratings. Table 4 also shows that sets of unordered categories were used as rating tasks in 18 articles. For example, in a study on punishment norms (Jacoby and Cullen, 1998), respondents were asked to choose one or several punishment types for the offenders described in the vignettes. Further, quite a few articles made use of the form of rating task identified as an amount, for example an amount of money (e.g. Shepelak and Alwin, 1986). In five articles, respondents were asked to state the probability, on a scale between 0 and 100, that either they themselves or some other person would act in a certain way (e.g. Applegate et al., 2000).

In order to make a distinction between studies using single rating tasks as constructs, and those combining multiple items for measuring a particular judgement, I chose to make a special category for the latter form of practice. Included in this category is the article by Phelan et al. (1997), for example, in which respondents were asked to rate descriptions of a poor man on 16 items with four ordered response options each. A principal-components factor analysis of the responses to the items resulted in a number of scales (dimensions), which were subsequently used as dependent variables in the analyses. Most factorial survey studies make use of rating scales that may be employed in advanced statistical analysis. However, a number of articles included in this review employed open-ended rating tasks, where respondents were asked to give a rationale for their judgements of the vignettes. All of these articles were written by the same team of authors (e.g. Ganong and Coleman,

[13] Naturally, variables consisting in respondents' ratings of vignettes may be employed as predictors of other judgements (e.g. Opp, 2002).
[14] Only rating tasks relating to the judgement(s) of interest in the article were counted. In addition, each type of rating task was counted only once for each article.

Table 5
Vignette sampling (n = 92).

	Frequency
Vignette universe	22
Vignette sample	
Simple random sampling	
With replacement	44
Without replacement	1
Not further specified	6
Clustered random sampling	10
Stratified sampling	3
Fractional factorial sampling	1
Other sampling technique	2
No information	3

2006a), who employed an "extended" version of the FSA design, designated the "multiple segment factorial vignette design" (Ganong and Coleman, 2006b, p. 455).[15] Finally, among the 106 articles reviewed in this article, five made use of rating tasks that did not fit any of the above-described categories. For example, Buskens and Weesie (2000) used the paired comparison method, which involved asking their respondents to make 10 choices between pairs of scenarios describing circumstances surrounding the sale of a used car.

3.2. Vignette designs in factorial surveys

In factorial surveys, the composition of the vignettes specifies the determinants of the judgements (other than those relating to the respondents) that may be studied and also determines how well they may be studied. According to Jasso, the selection of dimensions for the vignettes ought to be guided by "prior theory and research, extra-theoretical reasonings, and conventional wisdom" (Jasso, 2006, p. 342). A similarly careful approach must also be used to determine the choice of levels for the dimensions, since it will only be possible to measure any effect of a given dimension on the judgement under investigation if the dimension levels chosen accurately capture potential variation in respondents' judgements. The question of the number of dimensions to select for a factorial survey must naturally be considered in relation to the objective of the study. It should be noted however that the inclusion of additional dimensions results in lengthier descriptions that must then be read by the respondents, which may increase the risk for respondent fatigue and thus result in lower levels of reliability and validity in the findings (cf. Batista-Foguet et al., 1990). Further, when deciding on the number of levels to include for each dimension, the researcher needs to take into consideration how many predictors may be included in the statistical models, given the number of vignettes that are to be judged by the respondents. As has been pointed out by Jasso (2006), quantitative dimensions (continuous variables) may include any number of levels, while the number of levels to include for qualitative dimensions (discrete variables) may need to be restricted, since all but one of these levels will be included as a separate predictor in the statistical model (with the use of dummy coding).[16] The number of dimensions in the 92 studies reviewed varied between 2 (n = 5; e.g. Steen and Cohen, 2004) and 25 (Thurman et al., 1988), with the median number lying at six dimensions. Sixteen of the vignette designs included less than four dimensions, with only four studies being based on designs with more than 15 dimensions. As for the dimension levels, the number of levels in the designs varied between 1[17] (n = 2; e.g. Herzog, 2005b) and 110 (Bose and Rossi, 1983). Actually, the dimensions consisting of only one fixed level also included a so-called "blank" level (these levels were not included in the calculations). Blank dimension levels are used with the specific purpose of leaving out information about one or several dimensions from a number of vignettes in the design (Rossi and Anderson, 1982). Although far from all authors specified the number of vignettes making up the full vignette universe, it should be noted that at least a few designs comprised only four possible combinations of dimension levels (e.g. Phelan et al., 1997) whereas Jasso (1998) explicitly reported that the vignette universe in her data, which had been collected by Rossi et al. (1985), contained 1,047,259,295,424 different vignettes.

As was mentioned in the introduction, one of the main characteristics of factorial surveys is that each respondent judges only a subset of the fictive descriptions constituting the vignette universe. If the vignette universe is small, this may mean that all combinations of dimension levels are judged at least once. As can be seen from Table 5, this was the case for 22 of the studies reviewed in this article (e.g. Corrigan et al., 2006). However, since most vignette universes were far too large to allow for judging all possible combinations, it was more common to draw a probability or a systematic sample of fictive descriptions. Table 5 shows that by far the most common sampling strategy in the FSA designs reviewed was simple random

[15] Another feature of this design is that the respondents are presented with a series of vignettes that develop over time (see e.g. Ganong and Coleman, 2006b).
[16] Rossi and Anderson (1982) suggested that authors of factorial surveys may use a technique referred to as "coding proportional to effect" for transforming a discrete variable (constituted of dummy variables for each of a number of dimension levels) into a continuous variable (see pages 63-64).
[17] Some designs included one-level dimensions that were used only as control variables in the analyses (e.g. Steen and Cohen, 2004). These dimensions and levels are not included in the calculations.

sampling with replacement (e.g. Killian and Ganong, 2002), which involves presenting a unique, randomly drawn, sample of vignettes to each respondent. A number of studies used a clustered random design (e.g. Jasso and Webster, 1997), randomly drawing a pre-specified number of unique samples of vignettes, each of which was judged by multiple respondents. In only one study (Rooks et al., 2000), was it explicitly stated that the authors used a simple random sampling design without replacement. This latter strategy is preferable when the researcher wishes to eliminate the risk of presenting identical vignettes to respondents (Dülmer, 2007).

When all levels of the same dimension are equally likely to be present in a vignette sample, as is the case in all of the sampling forms described above, the sample of vignettes may be described as rectangular (Rossi and Anderson, 1982) or, more commonly, symmetrical (Dülmer, 2007). One result of this is that the distribution of characteristics among the actors described in a sample of scenarios may differ from that of a real-world population, which commonly consists of subgroups of different sizes. An alternative but seldom used strategy (see Table 5) for improving the similarity between the vignette sample and a real-world population is to draw a stratified sample of vignettes. In a study of public views on the sentencing of juvenile murderers, for example, Applegate and Davis (2006) weighted certain dimension levels, so as to make their sample reflect the proportion of juveniles referred to juvenile courts in Florida in 2001. The fractional factorial sampling design, which involves the systematic selection of dimension level combinations that together form a sample that is both symmetrical (although sometimes this criteria is relaxed) and orthogonal (Dülmer, 2007), was employed in only one of the studies reviewed (Pampel et al., 1984).

In the analysis of data produced by a factorial survey, the vignette typically makes up the unit of analysis. The statistical power of the analyses is therefore affected both by the number of respondents taking part in the study and by the number of vignettes judged by each respondent.[18] Since respondents commonly judge multiple vignettes (see below), the authors of factorial surveys do not need to sample as many respondents as is the case in general social survey research. In fact one of the studies mentioned in the introduction of this article (Jasso, 1988) made use of only 12 professionals, each of whom judged 40 vignettes. The maximum number of respondents employed in any one of the presently reviewed studies was found in a study focused on a probability sample of 7021 U.S. residents (Wagenaar et al., 2001). As regards the average number of vignettes judged by respondents, the median was eight (e.g. Miller and Bukva, 2001) whereas the minimum and modal numbers of vignettes were both one ($n = 22$; e.g. Rindfleisch and Bean, 1988).[19] The maximum number of vignettes judged by a single respondent was 110 (Bose and Rossi, 1983).[20]

Before concluding this section, I would like to draw attention to one of the critical issues brought up by Rossi and Anderson (1982) in their introduction to the factorial survey approach, namely that of the realism—or external validity—of results generated by a factorial survey study. It is occasionally argued that it is important that the judgements explored by the method are "within the competence of the target population" (Wagenaar et al., 2001, p. 347). This does not necessarily mean that the respondents must have experienced situations similar to those described in the scenarios. They may instead have acquired ideas about such situations from the media, for example (Seron et al., 2004). More commonly, however, discussions of the external validity of the results from FSA studies focus on what if any distorting effects the approximate orthogonality of the vignette samples may have on respondents' judgements. As has been noted above, the fact that the vignettes are as a rule composed of uncorrelated vignette dimensions means that some respondents may be asked to judge scenarios that very rarely occur in the real world. A fairly common strategy for reducing the number of "unrealistic" vignettes in the sample is to impose restrictions as to which combinations of dimension levels are allowed to occur ($n = 20$). When such restrictions are imposed, correlations between dimensions are built into the design. In the studies reviewed, a substantial majority of the restrictions made related to combinations that were not only "uncommon", but also logically impossible. In a study about the rationality of sexual offending, for example, where male students were asked to assign their own risk perceptions to the circumstances described in the scenarios, Bachman et al. (1992) chose to exclude vignettes in which the female victim had received cuts and bruises from the male perpetrator even though he had used no physical force. In addition to the purely analytical drawbacks that are generated when correlations between dimensions are included in the design, other problems may also crop up. Hunter and McClelland (1991) suggest for example that any restrictions imposed on a vignette design that reduce the number of possible combinations of dimension levels may make the vignettes judged by respondents more repetitive, and thus also less plausible. A solution to this problem has been provided by Shively (2001), who chose to include several additional dimensions in the vignettes, that were not then considered as explanatory variables, in order to provide more changeable and complex (and thus more realistic) scenarios. Finally, in order to "check" the realism of the vignettes, some authors have chosen to include an additional question in the questionnaire, asking the respondents to assess the plausibility of each vignette. This made it possible for the authors to evaluate which combinations of dimension levels were judged to be unrealistic by respondents, and to drop these vignettes from subsequent analyses (e.g. Simpson and Piquero, 2002).

[18] When the analyses are based on a large number of vignettes, it is possible to "increase the number of factors in the vignettes and to investigate interaction effects, and it increases the precision of the obtained estimates" (Jasso and Opp, 1997, p. 952).

[19] The second mode was 10 (n = 12; e.g. Ward et al., 2002).

[20] Standardized vignettes introduced at the beginning of a series of vignettes, which are usually employed in order to decrease potential judgement biases from reading the initial vignette, and to be able to control for respondents' rating tendencies (e.g. Ward et al., 2002), are not included in these figures.

Table 6
The social and individual components of judgements (n = 106).

	Frequency Yes	Frequency No
The social component	91	15
The individual component		
Individual variation	12	94
Subgroup variation	87	19
Judgement thresholds	81	25
Judgement processes	50	56
Judgement errors	21	85

4. Factorial survey analyses

This section presents the central modelling strategies and analytical techniques employed by the authors of the 106 factorial survey studies. In order to make this part of the review manageable, and to avoid introducing biases resulting from variations in the amount of the analytical work that different authors actually present in their articles, I have decided to focus on those analyses whose results are displayed in tables or in figures.[21] As a result of this strategy, the review does not cover all of the analyses discussed in the articles, but it does include those that the authors regard as being of most significance for their respective studies.

I will begin by providing a few details about the general techniques used in the data analyses. The review shows that by far the most common choice of multivariate statistical method was regression analysis. The majority of articles (n = 77) presented results from linear regression analyses. Regression analysis was also employed in designs with binary (n = 20; e.g. Steen and Cohen, 2004), ordered (n = 4; e.g. Emerson et al., 2001), nominal (Ganong and Coleman, 1998a) and skewed (n = 3; e.g. Nagin and Paternoster, 1993) dependent variables. Analysis of variance, which is the technique most commonly associated with the factorial experiment (Rossi and Anderson, 1982), constituted the sole multivariate statistical method employed in 10 articles (e.g. Meeker and Elliott, 1987). Since regression analysis was the most frequently employed analytical technique, my examples of modelling strategies will draw on the various opportunities inherent in regression analysis (the figures presented in the tables below include all analyses presented in tables or in figures, irrespective of the type of technique employed).

As was mentioned in the introduction to this article, one of the central assumptions made by Rossi and Anderson (1982) was that judgements of social objects are socially and individually structured, which is to say that judgements are made up of a social and an individual component. That judgements are socially structured means that there is some level of agreement among people in a given society (or in a smaller group) as to the (combinations of) factors that it is important to take into consideration when making a specific judgement. In other words, the social component of a specific judgement consists in the socially agreed-upon principles of that judgement (Rossi and Anderson, 1982). That judgements are individually structured implies that "each individual tends towards consistency in his or her own judgements" and that he or she may be assumed to depart from the socially agreed-upon principles of judgement in a relatively consistent way (Rossi and Anderson, 1982, p. 17).

The modelling of the social component of judgements is relatively straightforward, and involves estimating the effects of vignette dimensions on the respondents' judgements. In a regression-analytical framework, this means that the judgements provided by respondents constitute the dependent variable in the model, with the vignette dimensions being included as independent variables. Table 6 shows that 91 of the 106 articles reviewed included tables and/or figures which presented analyses of the social component of judgements. While 89 of these articles modelled main effects of vignette dimensions on respondents' judgements, 32 articles modelled effects of interactions between vignette dimensions on judgements, most commonly either by including multiplicative interaction terms in the analysis (e.g. Ganong and Coleman, 2006a) or by presenting analyses of main effects of vignette dimensions on the judgements separately for different levels of other dimensions (e.g. Denk et al., 1997).[22]

In contrast to the straightforward modelling of the social component of judgements, there are multiple strategies for exploring the individual component. At the most basic level, it should be noted that Rossi and Anderson's (1982) definition of the individual component denoted both truly "individual" variation in judgements and variation between "subgroups" of respondents (constructed by grouping respondents with identical values on some relevant continuous or discreet variable). The current review shows that there has been very little interest in modelling between-respondent variation in judgements. In fact, while subgroup variation was explored in 87 articles, only 12 articles (e.g. Rauma, 1991) presented analyses of individual variation in judgements. Looking more closely at the subject of subgroup variation in judgements, we may begin by concluding that it can be modelled in a variety of ways (Rossi and Anderson, 1982, see also Byers and Zeller, 1998): First,

[21] Exceptions were made for two articles in which there were no tables or figures (Roberts and White, 1986; Cummings and Armenta, 2002) and for an article in which the majority of results were presented in the form of equations (Warr, 1990).

[22] In a few articles (e.g. Miller et al., 1986), the authors modelled the social component using bivariate statistics only. Actually, given that the vignette dimensions are approximately orthogonal to one another, the patterns produced by bivariate analysis will remain when controls are included for other vignette dimensions.

Table 7
Strategies for coping with intrarater correlation (n = 106).

	Frequency
Not applicable	30
Not discussed	35
Argues no problem	2
"Double-check" analyses	8
Robust standard errors	10
Fixed effects models	3
Respondent-level models	3
Specific method, e.g. "multilevel" analyses	12
Other	3

subgroups of individuals may vary in the average levels of judgements rendered, or in their "judgement thresholds". Second, they may also vary in the weights they assign to different dimensions when forming the judgement, that is in their "judgement processes". Third, subgroups of individuals may vary in the extent of the normal stochastic error term, that is in their "judgement errors".

Table 6 displays the number of articles in which these three forms of subgroup variation were examined. Among the authors using regression analysis, subgroup variation in judgement thresholds was modelled chiefly by including one or more variables describing the respondents as predictors in the analyses (e.g. O'Toole et al., 1999). Differences in judgement processes, which were investigated in detail in 50 articles, were modelled in a variety of ways. The most common strategy was to estimate regression models that included vignette dimensions as predictors separately for different subgroups of respondents (e.g. Will and Cochran, 1995). While quite a number of authors included interactions between vignette dimensions and respondent variables as predictors in their regression analyses (e.g. Bell and Forde, 1999), another reasonably common strategy was to present regression analyses with respondent variables as predictors separately for different levels of dimensions (e.g. Schreckengost, 1987). Finally, a small number of authors presented models in which respondent variables were included as predictors of the estimated slopes of vignette dimensions on judgements (e.g. Rauma, 1991). The modelling of subgroup variation in judgement errors was automatically made possible whenever regression models with vignette dimensions as predictors of judgements were estimated separately for different groups of respondents, provided of course that the coefficients of determination (R^2) were included in the tables (e.g. Shlay, 1986).[23] The comparison of the "coefficients of alienation" for different subgroups of respondents (Byers and Zeller, 1998, p. 75)—computed by subtracting the value of R^2 from 1—shows the degree of subgroup variation in the normal stochastic error. Naturally, by comparing the coefficients of determination, it is also possible to explore the extent to which the overall importance of the vignette dimensions as predictors of judgements varies between subgroups.

As has already been noted, in factorial surveys, respondents' ratings of multiple vignettes are typically pooled in the analyses, and the vignette as a rule constitutes the unit of analysis. This means that the aggregate judgements of vignettes may be affected by so-called intrarater correlation (Rossi and Anderson, 1982). When the observed judgements cannot be assumed to be stochastically independent, as is the case in the presence of intrarater correlation, one of the primary assumptions of regression analysis is violated, namely that of the statistical independence of errors (no autocorrelation; Lewis-Beck, 1980). Since these errors are essentially random with respect to the vignette dimensions, potential intrarater correlation does not affect the estimation of the regression coefficients for the vignette variables (cf. Hunter and McClelland, 1991). However, standard errors may be underestimated, resulting in an increased risk for incorrectly rejecting a true null hypothesis. Due to the inflated sample size, this problem is even more critical for the standard errors of potential respondent variables. Table 7 shows factorial survey analysts' different strategies for coping with intrarater correlation.[24] In addition to the 30 articles where this was not an issue, the review includes 35 articles in which the authors did not discuss the risk for intrarater correlation despite the potential presence of this form of error in their analyses. The fact that this subject matter has not been dealt with as a rule is somewhat surprising, particularly considering that Rossi and Anderson urged future authors of factorial survey studies to search for a solution to this potential problem as early as 1982. However, not all FSA researchers are convinced that intrarater correlation is necessarily problematic. Cochran and his colleagues (2003) argued for example that problems due to intrarater correlation are most pronounced when each respondent judges 30 or more vignettes. Furthermore, all the authors who performed "double-check" analyses in their articles—for example by repeating the analyses using a sample consisting of only one vignette per respondent (e.g. St. John and Heald-Moore, 1995)—came to the conclusion that the estimates produced in their initial regression analyses were indeed robust. Table 7 also shows however that authors of factorial survey articles have employed a variety of strategies for dealing with intrarater error, including substituting robust standard errors (e.g. the Huber-White heteroscedasticity-corrected standard errors) for the ordinary standard errors (e.g. Jasso and Webster, 1999), employing

[23] However, due to the many forms of information provided by such regression models, one cannot deduce from the tables alone that the authors explicitly intended to model subgroup variation in judgement error.

[24] Naturally, this analysis was based on all the information provided in the articles.

so-called "fixed effects" models, in which dummy variables, representing each respondent, are added to the analyses (e.g. Porter, 2001), and performing the data analyses at the level of the respondent (e.g. Pampel et al., 1984).

For researchers who are primarily interested in applying the modelling strategies described earlier in this section of the paper, any of the above-mentioned strategies for dealing with intrarater correlation is sufficient. However, as is revealed in Table 7, a small number of FSA authors have chosen to analyse their data using analytical techniques that are particularly suited to data characterised by what is referred to as a hierarchical structure, i.e. data where units at a specific level (the level of the vignette) are clustered within units at a higher level (the level of the respondent). Most of these analytical techniques, which go under a number of different designations including "hierarchical" regression analysis (e.g. Shively, 2001), "error components" regression (Hennessy, 1993) and "random effects" analysis (e.g. Simpson and Piquero, 2002), may be described as variants of "multilevel" analysis. Besides correcting the standard errors for the potential effects of intrarater correlation, multilevel analysis allows for a further decomposition of the unexplained variance (i.e. the error term) into what are referred to as "variance components" for each level of the analysis (Hox, 2002). At the level of the respondent, variance components may be specified for the intercept and for the slopes of the regression coefficients included at the lower level of the analysis (i.e. the vignette dimensions). When the respondent-level variance component for the intercept is statistically significant, this means that there is variance among the respondents in the average levels of their judgements, i.e. in their judgement thresholds. A significant level-two variance component for the regression slope of a specific vignette dimension denotes the existence of respondent variation in the weight given to that particular dimension, i.e. in the judgement process. In order to make full use of the possibilities afforded by multilevel analysis, the level-two variance components should be explored prior to as well as subsequent to the inclusion of potential respondent predictors. By specifying what is referred to as a null model, for example, Degenholtz and his colleagues (1999) detected significant variation between case managers in their decisions as to whether or not to recommend an out-of-home placement for elderly clients.[25] More specifically, the case managers differed in both their judgement thresholds and their judgement processes related to the vignette dimension representing client preferences. Further analyses showed that part of the variation found in judgement thresholds and judgement processes could be explained by variables relating to the case managers (for more information, see Degenholtz et al., 1999). In sum, by using multilevel techniques, it is possible to model individual variation in judgement thresholds and judgement processes even without prior knowledge of what factors, if any, might explain this variation. In addition, by examining the variance components subsequent to the inclusion of respondent predictors, it is possible to conclude whether there remains any between-respondent variance to be explained (for an extensive discussion of why and how to combine factorial survey data and multilevel analysis, see Hox et al., 1991).

While most authors of factorial survey articles begin by estimating the socially agreed-upon principles of judgements and then continue by modelling subgroup and individual deviations from these principles, Guillermina Jasso has argued that the analyses should be carried out the other way round. Thus FSA researchers should begin by estimating the "equation-inside-the-head" for each respondent (Jasso, 2006, p. 338).[26] Subsequently, the researchers should compare different models of judgements, thus testing whether the respondents may be characterised by a single intercept and by the same slopes, or whether they are better represented by different intercepts and slopes. Six of the articles included in the review followed this analytical procedure (e.g. Hechter et al., 1999; for a thorough description, see Jasso, 2006).[27]

5. Concluding remarks

The objective of this article has been to provide an easy-to-follow overview of the various ways in which the factorial survey approach has been employed in sociology between 1982 and 2006. To this end, a review was conducted of 106 articles published in central sociology journals. This final section will outline the most important findings. A first general conclusion is that, over the years, the factorial survey approach has appeared only infrequently in the journals covered by this review. In fact, the average yearly publication rate is only slightly higher than four articles per annum. Further, an examination of the subdiscipline affiliations of these articles showed that nearly half of them dealt with issues relating to crime, law and deviance. While the approach can evidently be employed for the study of a wide variety of forms of judgements, its suitability for investigating norms may partly explain why it has so often been used in the study of judgements relating to unlawful behaviour. Moreover, the FSA articles published in these journals are not only concentrated within a particular subdiscipline of sociology, but also within a particular country (the United States). In addition, as can be seen from the list of the publications covered by this article (see list of references below), many authors had published several different articles (Guillermina Jasso, for example, was the sole or co-author of no less than seven articles). Thus, the evidence from the review suggests a highly concentrated use of the factorial survey approach within a fairly limited circle of researchers. This cannot, I believe, be attributed to limitations associated with the approach itself, but rather to the poor circulation of knowledge about its existence and the opportunities it affords. As has already been mentioned, other than the introductory text from 1982 (Rossi and Nock), and a selected number of advanced methodological articles published in sociology journals (e.g. Jasso,

[25] Due to the nesting of case managers within workplaces, Degenholtz and colleagues (1999) specified a third level of the model. The analyses also showed significant "contextual" variation in judgement thresholds.

[26] This requires a sufficient number of judgements made by each respondent.

[27] In fact, a large majority of these studies found that the respondents were indeed better represented by different intercepts and slopes (heterogeneity) than by the same intercept and the same slopes (homogeneity).

2006; Dülmer, 2007), the articles reviewed in this paper together constitute just about the only source of guidance for sociologists interested in applying the factorial survey approach to the study of human judgement.

Nonetheless, my inspection of 106 FSA articles shows that this approach has been used to study a variety of forms of judgements, including normative judgements, positive beliefs and individuals' estimations of their own actions, thoughts and feelings. While most studies were concerned with judgements made by the general population, a fairly large group of articles presented findings from studies of professional judgements. Due to the recent publication of articles with the specific purpose of introducing the factorial survey approach to the study of professional judgements in nursing (Lauder, 2002; Ludwick et al., 2004) and social work (Taylor, 2006), for example, we may expect an increased use of FSA in this particular field of research. In fact, this approach constitutes a very suitable instrument both for comparing members of different professions, and for comparing professions (as well as any other population) across countries (cf. Ludwick et al., 2004). In addition, and as has been noted by Rossi and Anderson (1982), the strategy of estimating respondent-specific models (cf. Jasso, 1988) may also be of benefit for "counsellors concerned with providing guidance" (Rossi and Anderson, 1982, p. 54; cf. Jasso, 2006).

As regards the design of vignette studies, this review has revealed a large variation in the number of dimensions and levels chosen for the vignettes. This serves to prove that the approach is indeed very flexible and may be adapted to a number of objectives. In addition, even though the form of rating task most frequently used by the authors of the studies reviewed was the ordered category scale, many other forms of rating tasks were also represented in the studies. When it comes to the sampling of vignettes from the theoretical vignette universe, the review shows that simple random sampling with replacement has been by far the most commonly used sampling strategy. However, it should be noted that alternative sampling strategies, such as the fractional factorial design, for example, have recently been described in favourable terms by one author (Dülmer, 2007) who questions the dominance of the random sampling of vignettes in factorial survey studies.

As far as the modelling of factorial survey data is concerned, the review shows that even though most FSA analysts apparently considered the modelling of the social component of judgements as central to their study, quite a number of them also focused on modelling subgroup variation in judgement thresholds and judgement processes. These models were commonly estimated using regression analysis. It should be noted, however, that only a few of the authors made use of analytical techniques that are particularly suited to hierarchically structured data. Since multilevel analysis not only corrects for potential intrarater correlation, but also provides various means of exploring consensus and variation in human judgements—the subject matter of factorial surveys—future authors should definitely consider employing this collection of techniques (cf. Hox et al., 1991; Jasso, 2006).

In the introduction to this article, it was stated that one of the most important undertakings for social scientists is to determine the social causes and the social outcomes of human judgements. The analysis section of this review article only presented modelling strategies that had been used in order to detect determinants of judgements. However, among the articles reviewed, there was one study that investigated whether internalized norms about giving behaviour (as measured by the respondents' ratings of vignettes) could predict actual giving behaviour (Horne, 2003). Jasso has further suggested that by incorporating an FSA design into longitudinal research, it would be possible not only to isolate the determinants of judgements, but also to specify "the effects of the components of beliefs/judgments on many important behaviors and outcomes at different points in the life course" (Jasso, 2006, p. 411). This would certainly be an interesting development in the future employment of the factorial survey approach.

To conclude, in a chapter included in a volume published to the honour of Paul Lazarsfeld, Rossi (1979) praised his former supervisor for articulating the ideas that subsequently formed the basis for the development of the factorial survey approach. In this chapter, Rossi wrote: "it has been a long time since Lazarsfeld scribbled his comments on the margins of the draft of my dissertation (Rossi, 1951). And it will be some time before the vignette technique becomes entrenched as a common tool in social research" (Rossi, 1979, p. 185). Hopefully, this moment is approaching.[28]

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[28] In memoriam of the principal authors of "Measuring Social Judgements", who are no longer with us. Andy Anderson died in 2001, Peter Rossi died in 2006 and Steven Nock died in 2008.

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